

## Midterm Exam: Help Sheet

ECON 441: Introduction to Mathematical Economics

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### Linear Algebra

The minor of the element  $a_{ij}$ , denoted by  $|M_{ij}|$  is obtained by deleting the  $i$ th row and  $j$ th column of the matrix and taking the determinant of the resulting matrix.

Whereas, cofactor  $|C_{ij}|$  is defined as:

$$|C_{ij}| = (-1)^{i+j} |M_{ij}|$$

Determinant for an  $n \times n$  matrix when expanding with respect to the first row is given by:

$$|A| = \sum_{j=1}^n a_{1j} |C_{1j}|$$

To find the inverse of matrix  $A$  take the transpose of its cofactor matrix  $C = [|C_{ij}|]$  to find the adjoint of  $A$  and divide it by the determinant of  $A$ .

$$A^{-1} = \frac{1}{|A|} \text{adj} A$$

Adjoint of a  $n \times n$  matrix

$$\text{adj} A = C' = \begin{bmatrix} |C_{11}| & |C_{21}| & \dots & |C_{n1}| \\ |C_{12}| & |C_{22}| & \dots & |C_{n2}| \\ \vdots & \vdots & \dots & \vdots \\ |C_{1n}| & |C_{2n}| & \dots & |C_{nn}| \end{bmatrix}$$

Cramer's rule: Given  $Ax = b$ , form matrix  $A_k$  by interchanging the  $k^{\text{th}}$  column of  $A$  by  $b$ , then

$$x_k^* = \frac{|A_k|}{|A|}$$